

All About Iron

What is Iron and Why Do We Need It?

Iron is a mineral that is found in many foods. Our body relies on iron to carry out essential functions including:

- Iron helps transport oxygen around the body in the blood.
- Iron helps cells release energy, which improves our energy levels.
- Iron helps produce red blood cells.
- Iron is required for a healthy immune system.

Approximate Daily Iron Requirements

Children	8-10 mg/day
Adolescents Females	15 mg/day
Adolescents Males	11 mg/day
Adult Females	18 mg/day
Adult Males	7 mg/day
Pregnant Females	27 mg/day

These amounts are approximate only, and will vary between individuals. Of note, women typically require much more iron intake than men do.

Sources of Iron

The body doesn't produce iron so we have to get it through the food we eat. Unfortunately, many of us don't get enough iron in our diet. In fact, iron deficiency is the world's most common nutritional deficiency!

We can get 2 types of iron from food:

- Haem iron: from animal sources
- Non-haem iron: from plant-based sources

Absorption is Key

It's not enough just to eat foods with a high iron content—that iron also needs to be absorbed! Haem iron is absorbed much more efficiently than non-haem iron. In fact, we absorb about 18-22% of the iron from animal sources, whereas we absorb less than 5% of the iron from plant-based sources. These types of iron are also absorbed differently so it can be helpful to be aware of what foods can increase or inhibit absorption.

Foods That Increase Iron Absorption

- **Animal Protein:** Animal protein such as meat, chicken, fish can enhance the absorption of non-haem iron
- **Vitamin C:** Berries, citrus fruits, tomato, and broccoli are high in vitamin C, which support iron absorption
- **Vitamin A:** Vitamin A also increases iron absorption, and is found in spreads, oils and full cream dairy products.

Foods That Decrease Iron Absorption

Some compounds or molecules in foods can actually *interfere* with iron absorption. For example:

- Compounds found in spinach, wholegrains, brans, and legumes (soy beans, split peas, lentils and dried beans) have been found to inhibit or reduce iron absorption
- Some molecules (called tannins) found in tea, coffee

and cola drinks also reduce iron absorption.

What Increases the Risk of Iron Deficiency?

- Low energy intake e.g., a low calorie diet
- Not eating enough iron rich food
- A poorly balanced vegetarian or vegan diet

Also, women who are pregnant or have regular periods are generally more likely to be iron deficient since their iron requirements are much higher.

Symptoms of Iron Deficiency

If you're not getting enough iron in your diet, you may experience tiredness, poor concentration, dizziness, muscle weakness and headaches. You are also at increased risk of infection.

How to Make Sure You Get Enough Iron

- Ensure you are eating enough (in terms of your total energy intake) and a balanced variety of foods.
- Choose breakfast cereals that contain added iron.
- Choose red meat 3 to 4 times per week. Small amounts in sandwiches, stir fry and soups can be useful if you prefer not to eat large pieces of meat.
- Add foods rich in Vitamins C and A to your meals.
- Limit intake of foods that inhibit iron absorption.
- Avoid drinking strong tea and coffee at meal times.

HAEM IRON SOURCE	Iron content (mg)	Iron Absorbed (mg)
Beef Liver (90g)	5.7	0.64
Sirloin Steak (90g)	2.8	0.42
Lamb Chop (90g)	1.8	0.27
Pork Tenderloin (90g)	1.4	0.16
Chicken Breast (90g)	0.9	0.13
White Fish (90g)	0.4	0.04
NON-HAEM IRON SOURCE	Iron content (mg)	Iron Absorbed (mg)
Bran Flakes (45g)	1.2	0.05
Kidney Beans (70g)	1.1	0.06
Tofu	1.7	0.09
Spinach	0.8	0.04
Raisins	0.4	0.02

Should I Take an Iron Supplement?

Supplementation (under the guidance of your medical practitioner) may be required to treat iron deficiency. Full recovery from iron deficiency can be slow and may take several months. Supplementation may be stopped or reduced when iron levels return to usual ranges. If you would like further advice on ensuring you have enough iron in your diet, we recommend you speak to a dietitian; preferably one who has experience working with eating disorders.

